

Standard Model Particles from GF(27)* — all quantum numbers algebraically forced [B]

FFGFT · J. Pascher 2026 · Doc. 346/347/348/349

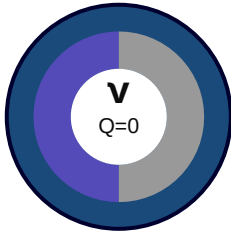
Generation 1

Generation 2

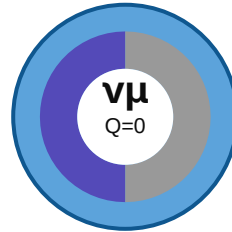
Generation 3

Neutrinos
(QR)

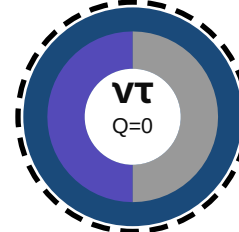
YL=-1
YR=—



YL=-1
YR=—

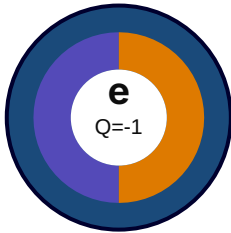


YL=-1
YR=—

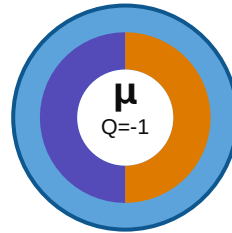


Charged
leptons
(QR)

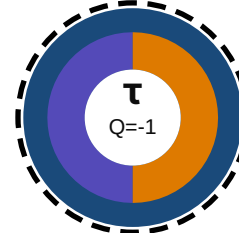
YL=-1
YR=-2



YL=-1
YR=-2

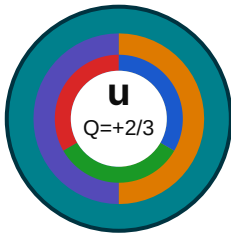


YL=-1
YR=-2

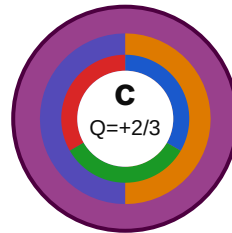


Up quarks
(NQR)

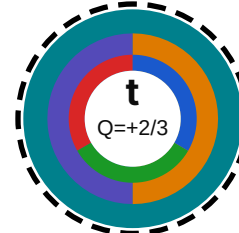
YL=+1/3
YR=+4/3



YL=+1/3
YR=+4/3

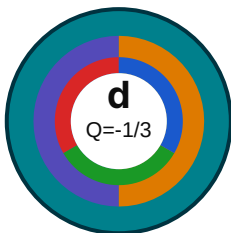


YL=+1/3
YR=+4/3

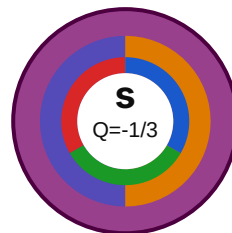


Down
quarks
(NQR)

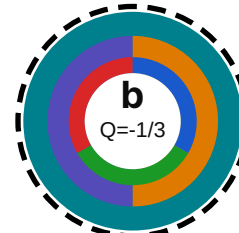
YL=+1/3
YR=-2/3



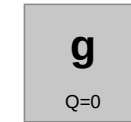
YL=+1/3
YR=-2/3



YL=+1/3
YR=-2/3



Gauge bosons



GF(3) fixed field [B] Doc.339



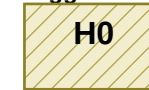
Gal(GF(9)/GF(3))=Z2 [K] Doc.336

Gluons gx8



4 Z13-Orbits x 2 Z2-par. = 8 [B]

Higgs



Q=0

T~.m=1 replaces Higgs (Doc.019) Exp. established [X]

Legend

Outer ring = Galois orbit:

- O1/O3: Gen.1/3 leptons (blue)
- O3: Gen.2 leptons (light blue)
- O2: Gen.1/3 quarks (teal)
- O4: Gen.2 quarks (berry)

Middle ring = chirality:

- Left violet = Sd = left-h. [B]
- Right orange = Su = right-h. [B]
- Gray = vR absent; Dirac [B/S]

Colour ring (quarks only):

- r/g/b = colour charge [B] Doc.346

Red dashed = SU(2)L doublet

Black dashed = Gen.3

[B] algebraically proved

[K] numerically [X] experimentally